Summary in Graph

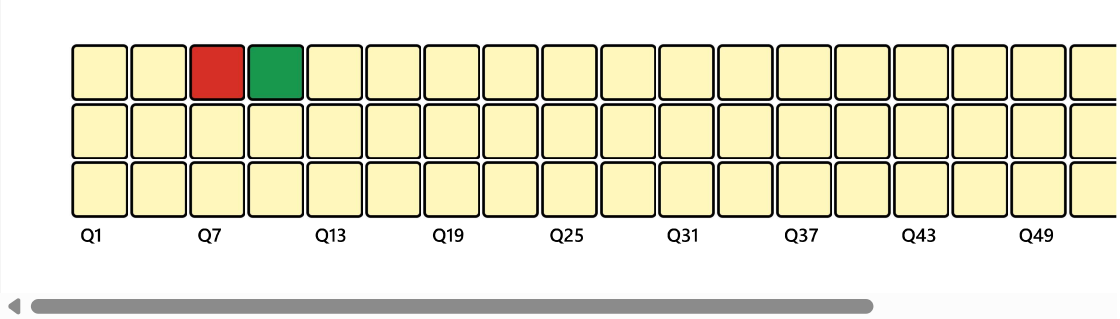
Exam Summary (GO Classes DA Test Series 2026 | MOCK GATE | Test 5)

Qs. Attempted:	2 (0 + 2)	Correct Marks:	2 (0 + 2)
Correct Attempts:	1 (0 + 1)	Penalty Marks:	0.67 (0 + 0.67)
Incorrect Attempts:	1 (0 + 1)	Resultant Marks:	1.33 (0 + 1.33)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	27 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (17) out of 167 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Granny gave away her entire collection of antique spoons to three people. Her daughter received 8 more than a third of the total; her son received 8 more than a third of what was then left; finally her neighbour received 8 more than a third of what was then left.

What is the sum of the digits of the number of spoons which were in Granny's collection?

A. 14
B. 12
C. 10
D. 8

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The five integers $2, 5, 6, 9, 14$ are arranged into a different order. In the new arrangement, the sum of the first three integers is equal to the sum of the last three integers. What is the middle number in the new arrangement?

A. 2
B. 5
C. 6
D. 14

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

When Ellen went to the shop, she found she could spend all her money on 6 cans of cola and 7 croissants or on 8 cans of cola and 4 croissants. If she decided to buy only croissants, how many croissants could she buy?

A. 12
B. 13
C. 15
D. 16

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

In the grid below each of the blank squares and the square marked X are to be filled by the mean of the two numbers in its adjacent squares. Which number should go in the square marked X?

10			X		25

A. 15
B. 16
C. 17
D. 19

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 26min 46sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

The movie was funny and I _____.

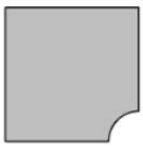
A. could help laughing

B. couldnt help laughed
C. couldnt help laughing
D. could helped laughed

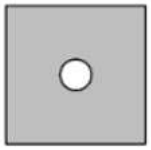
Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

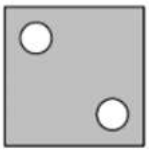
Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

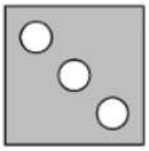
A square of paper is folded in half four times to obtain a smaller square. Then a corner is removed as shown.

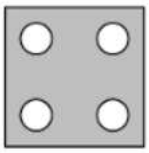


Which of the following could be the paper after it is unfolded?

A. 

B. 

C. 

D. 

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A drawer contains ten identical yellow socks, eight identical blue socks and four identical pink socks. Amrita picks socks from the drawer without looking. What is the smallest number of socks she must pick to be sure that she has at least two pairs of matching socks?

A. 5

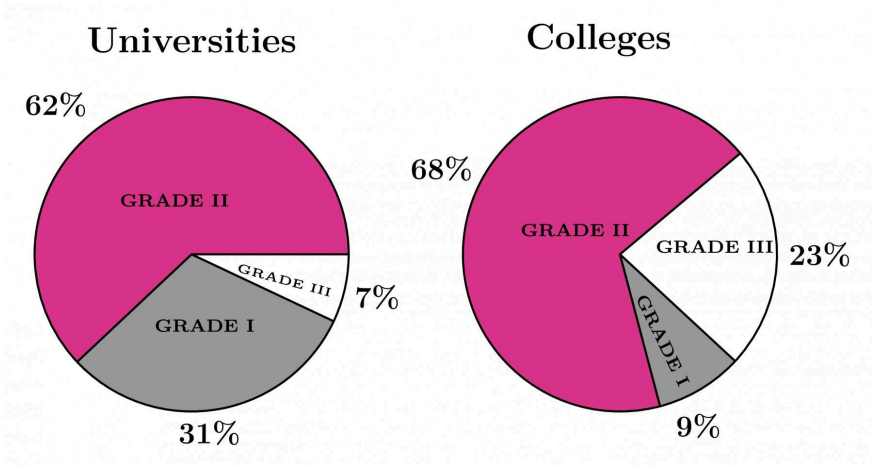
B. 6
C. 8
D. 11

Your Answer: A | Correct Answer: B | Incorrect | Time taken: 00min 23sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A certain country has 504 universities and 25951 colleges. These are categorised into Grades I, II, and III as shown in the given pie charts.

What is the percentage, correct to one decimal place, of higher education institutions (colleges and universities) that fall into Grade III?



A. 22.7
B. 23.7
C. 15.0
D. 66.8

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

If x satisfies the equation $4^{8^x} = 256$, then x is equal to _____.

A. $\frac{1}{2}$
B. $\log_{16} 8$
C. $\frac{2}{3}$
D. $\log_4 8$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Verbal Aptitude

Under a certain legal system, prisoners are allowed to make one statement. If their statements turns out to be true then they are hanged. If the statements turns to be false then they are shot. One prisoner made a statement and the judge had no option but to set him free. Which one of the following could be that

statement?

A. I did not commit the crime
B. I committed the crime
C. I will be shot
D. You committed the crime

Your Answer: C | Correct Answer: C | Correct | Time taken: 00min 04sec | Discuss

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the following values that can be associated with a node N. Let S be the start node and G be the goal.

```
    A: number of hops from S to N
    B: 1 - A
    C: actual path cost from S to N
    D: estimated path cost from S to N
 5. E: actual path cost from N to G
    F: estimated path cost from N to G
    G: actual cost from N to its nearest neighbour
    H: number of hops from N to G
    I: C + E
10. J: C + F
    K: D + E
    L: D + F
    M: NONE (none of the above values are used)
```

Given that each search algorithm refines the candidate with the lowest value associated with each candidate, identify the value used by each of the following algorithms.

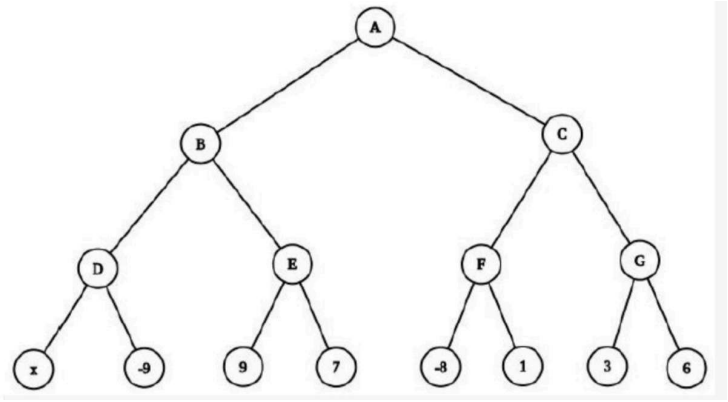
A. Depth First Search uses B
B. Breadth First Search uses A
C. Best First Search uses G
D. Algorithm A^* uses L

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Consider the following game tree.

Player 1 (max node) chooses the left action for which of the following value(s) of x.
Assume optimal play and Player 1 's turn at Node A.



A. 2
B. -7
C. 4
D. -1

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let b, y, a_1, a_2 be vectors in $\mathbb{R}^3$. Consider a plane P passing through y and spanned by a_1 and a_2. The point b is not on the plane P.

The distance between b and P is the minimum distance $\|b - y\|_2$. In which subspace of $A = (a_1 a_2)$ does $b - y$ lie?

A. Column Space $C(A)$
B. Row Space $R(A)$
C. Left Null Space $N\left(A^T\right)$
D. Null Space $N(A)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $\boldsymbol{A}$ be an $\boldsymbol{m} \times \boldsymbol{n}$ real matrix with SVD in standard outer product form $A = \sigma_1 \vec{u}_1 \vec{v}_1^{\top} + \sigma_2 \vec{u}_2 \vec{v}_2^{\top} + \sigma_3 \vec{u}_3 \vec{v}_3^{\top}$ with $\sigma_1 \geq \sigma_2 \geq \sigma_3 > 0$.

Which of the following is/are TRUE?

A. $A^{\top} A \vec{v}_2 = \sigma_2^2 \vec{v}_2$
B. $\operatorname{rank}\left(A^{\top}\right) = 3$
C. $\vec{v}_1 \vec{v}_1^{\top} = 1$
D. $[\,\vec{u}_1 \quad \vec{u}_2 \quad \vec{u}_3\,]^{\top} [\,\vec{u}_1 \quad \vec{u}_2 \quad \vec{u}_3\,] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Only the first row of the 4×4 matrix A is given to you:

$$A = \begin{bmatrix} 2 & 3 & -4 & 1 \\ ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix}$$

From your knowledge of what it means for a vector to be an eigenvector of A, if $[2, 0, 3, -4]^T$ is an eigenvector, the corresponding eigenvalue is _____.

Your Answer: | Correct Answer: -6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose $A \in \mathbb{R}^{n \times n}$ is a positive semi-definite (PSD) matrix. Which of the following is/are always true about A?

A. All eigenvalues of A are non-negative.
B. All elements of A are non-negative.
C. A is invertible.
D. $x^T A x \leq 0$ for all x.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X and Y be independent random variables with means μ_1, μ_2 and variances σ_1^2, σ_2^2, respectively. What is the correlation coefficient ρ between X and $Z = X - Y$ in terms of σ_1^2 and σ_2^2?

A. $\rho = \frac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2}$

B. $\rho = \frac{\sigma_1^2}{\sqrt{\sigma_1^2 + \sigma_2^2}}$

C. $\rho = \frac{\sigma_1}{\sqrt{\sigma_1^2 + \sigma_2^2}}$

D. $\rho = \frac{\sigma_1}{\sqrt{\sigma_1^2 - \sigma_2^2}}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 1 | Penalty: 0 | Probability

Let X and Y be independent random variables such that:

$$P(X = 1) = P(X = -1) = \frac{1}{2}, \quad P(Y = 1) = P(Y = -1) = \frac{1}{2}$$

Define $Z = X \cdot Y$, the product of X and Y. What is $P(Z = 1)$? Provide your answer up to 2 decimal places.

Your Answer: | Correct Answer: 0.50 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let A and B be two events. Suppose the probability that neither A nor B occurs is $2/3$. What is the probability that one or both events occur?

A. $3/4$
B. $2/3$
C. $1/2$
D. $1/3$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Consider some convex function $f : \mathbb{R}^d \to \mathbb{R}$, and assume that f is twice-differentiable. What can we say about $\nabla_x^2 f(x) \in \mathbb{R}^{d \times d}$, the Hessian of $f(x)$? (Hint: think about the case when $d = 1$, and what the second derivative of f says about its shape.)

A. $\nabla_x^2 f(x)$ is negative semi-definite.

B. $\nabla_x^2 f(x)$ is negative definite.

C. $\nabla_x^2 f(x)$ is positive definite.

D. $\nabla_x^2 f(x)$ is positive semi-definite.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Find the limit:

$$\lim_{x \to 1} \frac{x - 1}{\sqrt{2x + 23} - 5}$$

A. $\frac{5}{2}$

B. 5

C. $\frac{1}{2}$

D. 10

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are using a 1-Nearest Neighbor (1-NN) classifier with $\mathrm{L2}$ distance (Euclidean distance) to classify emails. The training set has $\mathbf{n}$ examples, and the validation set has $\mathbf{m}$ examples. Assume computing the $\mathrm{L2}$ distance between two emails takes constant time.

What is the asymptotic time complexity of computing the validation error?

A. $O(n + m)$
B. $\mathrm{O(nm)}$

C. $O\left(n^2+m^2\right)$
D. $O\left(n^2 m\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

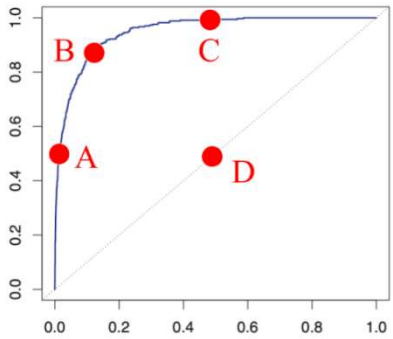
Which of the following properties is true about the (original) Perceptron algorithm?

A. The Perceptron always converges to the best linear separator for a given dataset.
B. The convergence criteria for Perceptron depends on the initial value of the weight vector.
C. If the dataset is not linearly separable, the Perceptron algorithm does not converge and keeps cycling between some sets of weights.
D. If the dataset is not lineary separable, the Perceptron algorithm learns the linear separator with least misclassifications.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You trained a classifier whose ROC curve appears below. You would like to correctly classify points that are in-class, but you don't care what labels you assign to points that are out-of-class. In other words, your (asymmetric) loss function is 1 when your prediction is $z=-1$ and the true label is $y=1$; otherwise, the loss is zero. The four red points below signify four classifiers. Of the four classifiers, which one has the lowest empirical risk on the test points?



A. Classifier A
B. Classifier B
C. Classifier C
D. Classifier D

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Considering single-link and complete-link hierarchical clustering, is it possible for a point to be closer to points in other clusters than to points in its own cluster? If so, in which approach will this tend to be observed?

A. No
B. Yes, single-link clustering
C. Yes, complete-link clustering
D. Yes, both single-link and complete-link clustering

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Select Type | Award: 1 | Penalty: 0 | Machine Learning

This question concerns PCA and LASSO. (Select all statements that are true)

A. PCA and LASSO can both be used for dimensionality reduction.
B. Lasso selects a subset (not necessarily a strict subset) of the original features
C. PCA and LASSSO both allow you to pre-specify how many features are chosen.
D. PCA produces features that are linear combinations of the original features.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Machine Learning

Suppose our input is two-dimensional sample points, with ten non-exclusive classes those points may belong to (i.e., a point can belong to more than one class). To train a classifier, we build a fully-connected neural network (with bias terms) that has a single hidden layer of twenty units and an output layer of ten units (one for each class). How many trainable parameters does this network have?

Your Answer: | Correct Answer: 270 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Assuming myLst = ["Basil", "Kookie", "KitKat", "Grey"], what does the expression myLst[:-2] evaluate to?

A.
```
["Basil, "Kookie"]
```

B.
```
["Basil, "Kookie", "KitKat"]
```

C.
```
["Kookie, "KitKat", "Grey"]
```

D. Error

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

We use the dictionary:

M	domain of men
s	Sharon
$N(x)$	x is nice
$L(x,y)$	x loves y

Which of the following formulas corresponds to the sentence:

All nice men love Sharon.

A. $\exists x \in M[N(x) \wedge L(x,s)]$
B. $\exists x \in M[N(x) \rightarrow L(x,s)]$

C. $\forall x \in M[N(x) \wedge L(x, s)]$
D. $\forall x \in M[N(x) \rightarrow L(x, s)]$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Artificial Intelligence

Let p, q be two atomic propositional assertions. Then which of the following is/are false?

A. $(p \rightarrow q) \vee (p \rightarrow \neg q)$ is a tautology.
B. $(p \rightarrow q) \vee (q \rightarrow p)$ is a tautology.
C. $(p \rightarrow q) \vee (q \rightarrow \neg p)$ is a tautology.
D. $(p \rightarrow q) \vee (\neg q \rightarrow \neg p)$ is a tautology.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Which of the following is/are True?

A. ER diagrams are used to model the physical schema of a database.
B. An entity set E can be associated with a relationship set R more than once.
C. At most 2 distinct entity sets can be associated with any relationship set R.
D. A weak entity set must have total participation in its identifying relationship set.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

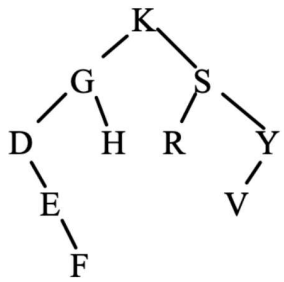
Q #22 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the attribute set $R = ABCDEF$ and the functional dependency set $F = \{AD \rightarrow B, A \rightarrow E, C \rightarrow E, DEF \rightarrow A, F \rightarrow D\}$. The number of superkeys in R is _____.

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following binary tree:



Which of the following is in-order traversal of the tree?

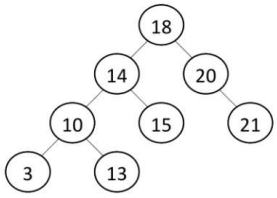
A. $\mathrm{E\ D\ F\ G\ H\ K\ R\ S\ V\ Y}$

B. D E F G H K R S V Y
C. E D F G H R K Y S V
D. D F E G K H R S Y V

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

The binary search tree shown below was constructed by inserting a sequence of items with integer keys into an empty tree.



Which of the following input sequences will produce this binary search tree?

A. 18 20 21 14 10 15 3 13
B. 18 14 20 10 15 21 3 13
C. 18 14 20 10 15 21 13 3
D. 18 15 29 10 14 3 13 21

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose X and Y are two independent random variables both following Poisson distribution with parameter λ. What is the value of $E(X-Y)^2$?

A. λ
B. 2λ
C. λ^2
D. $4\lambda^2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider these training examples:

$\mathbf{x}$	y
$(0,0)$	1
$(1,2)$	1
$(2,3)$	0
$(5,0)$	0

How does 3-NN (three nearest neighbors) classifiy $(2,1)$ using $\mathrm{L}1$ distance?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider two SVMs trained on a $2D$ feature space $\mathbf{x}$. The first SVM has a decision boundary defined as $\begin{bmatrix}3\\4\end{bmatrix}^T \mathbf{x} + 5 = 0$, while the second SVM has a boundary given by $\begin{bmatrix}5\\12\end{bmatrix}^T \mathbf{x} + 13 = 0$.

Which of the following statements is true about the margins of these SVMs?

Choose the correct answer.

A. The SVM with boundary $\begin{bmatrix}3\\4\end{bmatrix}^T \mathbf{x} + 5 = 0$ has a larger margin than the SVM with boundary $\begin{bmatrix}5\\12\end{bmatrix}^T \mathbf{x} + 13 = 0$.

B. The SVM with boundary $\begin{bmatrix}5\\12\end{bmatrix}^T \mathbf{x} + 13 = 0$ has a larger margin than the SVM with boundary $\begin{bmatrix}3\\4\end{bmatrix}^T \mathbf{x} + 5 = 0$.

C. Both SVMs have the same margin.

D. The margins of the SVMs cannot be determined from the given information.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We are doing binary classification on classes $\{1, 2\}$. We have a single dataset of size N of which a fraction α of the elements are in class 1. To construct a test set, we randomly choose a fraction β of the dataset to put in the test set, keeping the remaining elements in a training set.

We would like to avoid the situation where in the training or test sets, either class appears less than $0.1\ N$ times. In which of the following situations does this occur, in expectation?

A. $\alpha = 50\%, \beta = 50\%$
B. $\alpha = 40\%, \beta = 60\%$
C. $\alpha = 20\%, \beta = 70\%$
D. $\alpha = 30\%, \beta = 60\%$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider training a decision tree given a design matrix $X = \begin{bmatrix}6 & 3\\2 & 7\\9 & 6\\4 & 2\end{bmatrix}$ and labels $y = \begin{bmatrix}1\\0\\1\\0\end{bmatrix}$. Let f_1 denote feature 1, corresponding to the first column of X, and let f_2 denote feature 2, corresponding to the second column. Which of the following splits at the root node gives the highest information gain?

A. $f_1 > 2$
B. $f_2 > 3$
C. $f_1 > 4$
D. $f_2 > 6$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

In a binary search tree (BST), node A is a leaf, and node B is its parent. Which of the following statements is true regarding node B relative to node A?

A. B is either the smallest element larger than A or the largest element smaller than A.
B. B is always the smallest element larger than A.
C. B is always the largest element smaller than A.
D. B has no relation to the ordering of elements with respect to A.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #31 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

A rooted binary tree of height at most 10, in which all nodes contain *distinct elements*, has identical pre-order and post-order traversals. The *height of the tree* is defined as the number of edges on the longest path from the root to any leaf. What is the maximum number of nodes that such a tree can have?

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Which of the following is/are true?

A. Suppose T is a completely filled binary search tree with 7 nodes, the worst-case number of comparisons for a successful find operation in T is 3
B. Suppose T is a completely filled binary search tree with 7 nodes. the best case number of comparisons for a successful find operation in T is 1
C. Suppose T is a binary search tree with 7 nodes, only one of which is a leaf. suppose all keys are equally likely, the integer closest to the average case number of comparisons for a successful find operation in T is 4
D. Suppose T is a completely filled binary search tree with 7 nodes. suppose all keys are equally likely. The integer closest to the average case number of comparisons for a successful find operation in T is 2

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

The following sequence of arrays shows how a list gradually becomes sorted. A few intermediate steps of the sorting process are shown.

Original Array:	7 0 2 4 5 3 9 1
	0 7 2 4 5 3 9 1
	0 2 4 7 3 5 9 1
Final Sorted Array:	0 1 2 3 4 5 7 9

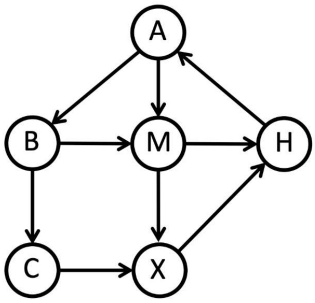
Which of the following sorting algorithms could produce this sequence?

A. Quick sort
B. Merge sort
C. Insertion sort
D. Selection sort

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the following directed graph.



There are several possible DFS orderings starting from node A. Consider the following DFS ordering:

$$\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{X}, \mathrm{H}, \mathrm{M}.$$

Which of the following statements is/are TRUE for the given DFS ordering?

A. The edge (M, X) is a cross edge.
B. The edge (M, H) is a cross edge.
C. The edge (H, A) is a back edge.
D. The edge (B, M) is a cross edge.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Select Type | Award: 2 | Penalty: 0 | Programming in Python

Given the following Python loop:

```
S = "hello"
for i in range(len(S)):
    print(S[i])
```

Which of the following options produces the same output as the original loop?

A.
```
for i in range(-len(S), 0): print(S[i])
```

B.
```
for i in range(-1, -len(S)-1, -1): print(S[i])
```

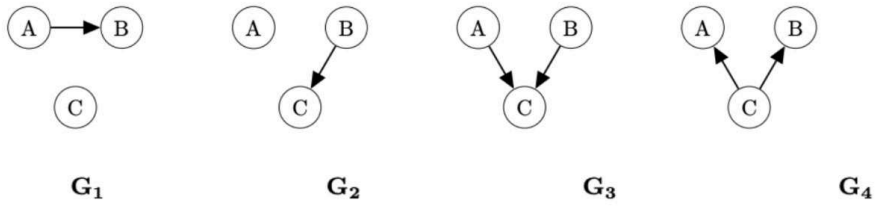
C.
```
for char in S: print(char)
```

D.
```
for i in range( 0 , len(S)): print(S[i])
```

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

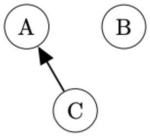
Q #36 | Multiple Select Type | Award: 2 | Penalty: 0 | Artificial Intelligence

Assume we are given the following four Bayes' nets, labeled $\mathbf{G}_1$ to $\mathbf{G}_4$:



$\mathbf{G_1}$ $\mathbf{G_2}$ $\mathbf{G_3}$ $\mathbf{G_4}$

Assume we know that a joint distribution $\mathbf{d}_1$ (over $\mathbf{A}, \mathbf{B}, \mathbf{C}$) can be represented by Bayes' net $\mathbf{B}_1$. Mark all of the following Bayes nets that are guaranteed to be able to represent d_1.



$\mathbf{B_1}$

A. G_1
B. G_2
C. G_3
D. G_4

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Find the distance between $\vec{w} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and the subspace $W = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$.

A. 1
B. $\sqrt{3}$
C. 2
D. $\sqrt{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given the matrix

$$A = \begin{bmatrix} -1 & 1 \\ -6 & 4 \\ 2 & -1 \end{bmatrix}$$

and vector

$$\mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$$

find the least squares solution to the equation $A\mathbf{x} = \mathbf{b}$. The least squares solution minimizes the squared error between $A\mathbf{x}$ and $\mathbf{b}$.

A. $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

B. $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

C. $\mathbf{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

D. $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Suppose that $T : \mathbf{R}^n \to \mathbf{R}^m$ is a linear transformation with standard matrix A.
Which one of the following statements is equivalent to the condition that T is one-to-one?

A. For each y in $\mathbf{R}^m$, the matrix equation $Ax = y$ is consistent.
B. For each x in $\mathbf{R}^n$, there is at most one y in $\mathbf{R}^m$ so that $T(x) = y$.
C. For each y in $\mathbf{R}^m$, there is at most one x in $\mathbf{R}^n$ so that $T(x) = y$.
D. A has a pivot in every row.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #40 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following is/are TRUE?

A. If A is a 3×5 matrix with 3 pivots, then the equation $Ax = b$ has at least one solution, regardless of what b is.
B. If A is a 5×3 matrix with 3 pivots, then the equation $Ax = b$ has at least one solution, regardless of what b is.
C. If $Ax = 0$ has infinitely many solutions, then $Ax = b$ has infinitely many solutions.
D. If $Ax = b$ has infinitely many solutions, then $Ax = 0$ has infinitely many solutions.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following 3×3 matrices are necessarily having 3 linearly independent eigen vectors?

A. A matrix with three distinct real eigenvalues.
B. A symmetric matrix with two real eigenvalues.
C. A matrix with a real eigenvalue $\lambda 1$ of algebraic multiplicity 2, such that corresponding to $\lambda 1$ there are 2 linearly independent eigenvectors.
D. A matrix with a real eigenvalue $\lambda 1$ such that corresponding to $\lambda 1$ there are 2 linearly independent eigenvectors.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following 3×5 matrix:

$$A = \begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ 0 & 2 & 5 & -1 & -3 \\ 1 & 2 & 3 & -1 & 0 \end{bmatrix}.$$

Which of the following statements is/are FALSE?

A. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix} \right\}$ is a basis of $\mathrm{Col}\, A$.

B. $\left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \\ -1 \end{bmatrix} \right\}$ is a basis of $\mathrm{Col}\, A$.

C. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 8 \\ 5 \\ 3 \end{bmatrix} \right\}$ is a basis of $\mathrm{Col}\, A$.

D. All statements are TRUE.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

With the SVD, we write $X = UDV^\top$. For which of the following matrices are the eigenvectors the columns of U?

A. $X^\top X$
B. $X^\top X X^\top X$
C. $XX^\top$
D. $XX^\top XX^\top$

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider the following joint probability distribution of random variables X and Y:

	$Y=0$	$Y=1$
$X=0$	$\frac{5}{12}$	$\frac{1}{12}$
$X=1$	$\frac{1}{3}$	$\frac{1}{6}$

Let $X=1$ represent having ice cream, $X=0$ represent not having ice cream, $Y=1$ represent getting sunburnt, and $Y=0$ represent not getting sunburnt.

Which of the following statement(s) is/are correct?

A. $P(X=0)=\frac{1}{2}$.

B. $P(Y=1)=\frac{1}{4}$.

C. X and Y are independent because $P(X,Y)=P(X)P(Y)$ for all values of X and Y.

D. X and Y are dependent.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

What is the expected number of times you need to toss a fair coin until you see both a heads and a tails at least once?

A. 1
B. 2
C. 3
D. 4

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X_1, X_2, X_3, X_4 be i.i.d. random variables each assuming the value 1 and -1 with probability $\frac{1}{2}$ each. Then, the probability that the matrix $\begin{pmatrix} X_1 & X_2 \\ X_3 & X_4 \end{pmatrix}$ is nonsingular equals _____.

A. $1/2$
B. $3/8$
C. $5/8$
D. $1/4$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

In the Taylor expansion of the function $f(x)=e^{x/2}$ about $x=3$, the coefficient of $(x-3)^5$ is _____.

A. $e^{3/2}\frac{1}{5!}$

B. $e^{3/2}\frac{1}{2^5 5!}$

C. $e^{-3/2}\frac{1}{2^5 5!}$

D. none of the above

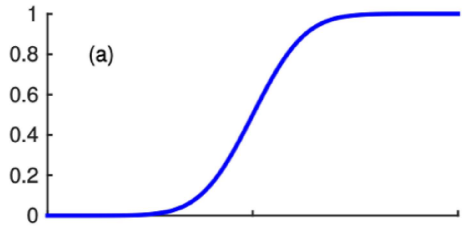
Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

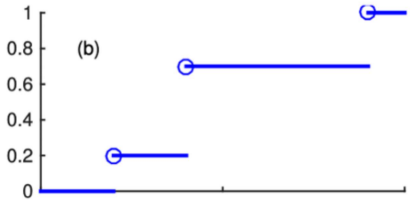
Consider the four graphs in the options.

Examine each graph and select which ones represent a valid Cumulative Distribution Function (CDF).

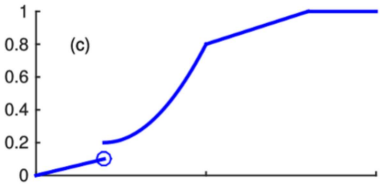
A.



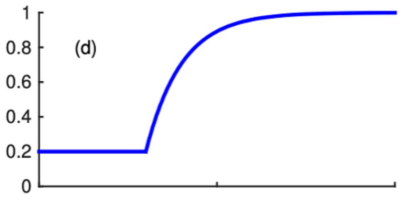
B.



C.



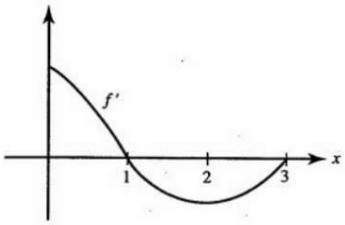
D.



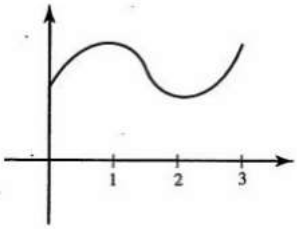
Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

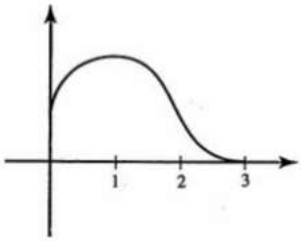
The graph of f'(the derivative of f) is shown below. What does the graph of f look like?



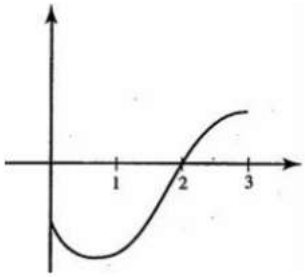
A.



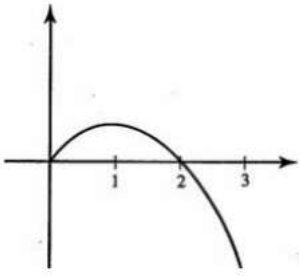
B.



C.



D.



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #50 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a binary classification $1D$ dataset with N examples that is linearly separable. A decision stump (a decision tree with only two leaves and depth 1) is used. The decision stump determines its threshold by finding the midpoint between the closest positive example and the closest negative example.

If the dataset contains at least two positive examples and two negative examples, what is the tightest upper bound on the leave-one-out error of the decision stump?

A. 0
B. $1/(2\ \mathrm{N})$
C. $1/\mathrm{N}$
D. $2/\mathrm{N}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

We have a function which takes a two-dimensional input $x = (x_1, x_2)$ and has two parameters $w = (w_1, w_2)$ given by $f(x, w) = \sigma\left(\sigma\left(x_1 w_1\right) w_2 + x_2\right)$ where $\sigma(x) = \dfrac{1}{1 + e^{-x}}$. We use backpropagation to estimate the right parameter values. We start by setting both the parameters to 0 . Assume that we are given a training point $x_1 = 1, x_2 = 0, y = 5$. Given this information answer the next two questions. What is the value of $\dfrac{\partial f}{\partial w_2}$?

A. 0.5
B. -0.25
C. 0.125
D. -0.5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following database tables:

```
    Books: bid INTEGER,
           title TEXT,
           library REFERENCES Library,
           genre TEXT,
 5.        PRIMARY KEY (bid)

    Library: lid INTEGER,
             lname TEXT,
             PRIMARY KEY (lid)
10.
    Checkouts: book INTEGER REFERENCES Books,
               day DATETIME,
               PRIMARY KEY (book, date)
```

We want to Return the bid and genre of each book that has ever been checked out. Remove any duplicate rows with the same bid and genre.

Which of the following SQL query is correct?

A.

```
SELECT b.bid, b.genre
FROM Books b, Checkouts c
WHERE b.bid = c.book
```

B.
```
SELECT b.bid, b.genre
FROM Books b, Checkouts c
WHERE b.library = c.library
```

C.
```
SELECT DISTINCT b.bid, b.genre
FROM Books b, Checkouts c
WHERE b.bid = c.book
```

D. None

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider two relations with the same schema: $R(A, B)$ and $S(A, B)$. Consider the following relational algebra expressions:

a. $\pi_{R.A}((R \cup S) - S)$
b. $\pi_{R.A}R - \pi_{R.A}(R \cap S)$
c. $\pi_{R.A}(R - S) \cap \pi_{R.A}R$

Which of the above relational algebra expressions is/are equivalent?

A. a, b
B. a, c
C. b, c
D. None

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a set of labeled examples $S = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ in $\mathbb{R}^n$. The soft-margin SVM formulation is given as:

$$\min_w \|w\|_2^2 + C\sum_{i=1}^{m}\xi_i, \quad \text{subject to} \quad y_i\left(w^T x_i\right) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Which of the following statements is TRUE about support vectors in this SVM formulation?

A. For all support vectors, $\xi_i > 0$.
B. For all support vectors, $\xi_i = 0$.
C. For some support vectors, $\xi_i = 0$ and for others, $\xi_i > 0$.
D. For all support vectors, $\xi_i > C$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

The given implementation of A^* graph search may be incorrect!

Select all of the problems that the bugs may cause when executing graph search.

Note that the fringe is a priority queue.

Nodes are inserted into the fringe using the standard key for A^*, namely $f = g + h$.

h is a consistent heuristic.

```
    function A* GRAPH SEARCH(problem)
        fringe ← an empty priority queue
        fringe ← INSERT(MAKE-NODE(INITIAL-STATE[problem]), fringe)
        closed ← an empty set
 5.     ADD INITIAL-STATE[problem] to closed
        loop
            if fringe is empty then
                return failure
            end if
10.         node ← REMOVE-FRONT(fringe)
            if GOAL-TEST(problem, STATE[node]) then
                return node
            end if
            for successor in GETSUCCESSORS(problem, STATE[node]) do
15.             if successor not in closed then
                    ADD successor to closed
                    fringe ← INSERT(MAKE-SUCCESSOR-NODE(successor, node), fringe)
                end if
            end for
20.     end loop
    end function
```

A. The GetSuccessors function could be called multiple times on the same state.
B. The algorithm is no longer complete.
C. The algorithm could return a suboptimal solution.
D. The implementation is incorrect, but none of the above problems will be caused.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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